```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, accuracy_score

# 1. Load your public dataset (e.g., KDD Cup / NSL-KDD CSV format)
# Replace 'network_traffic_data.csv' with your actual dataset path
df = pd.read_csv('network_traffic_data.csv')

# 2. Basic Preprocessing: Encode categorical columns (e.g., protocol_type, flag)
cat_cols = ['protocol_type', 'service', 'flag']  # adjust based on your dataset columns
for col in cat_cols:
    if col in df.columns:
        df[col] = LabelEncoder().fit_transform(df[col])

# Assume 'class' or 'target' is the label column (0 for normal, 1 for anomaly/attack)
X = df.drop(columns=['class', 'target'], errors='ignore')
y = df['class']  # or df['target']

# 3. Train/Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 4. Feature Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

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# 5. Train Scikit-learn Model (Random Forest)
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)

# 6. Evaluate
y_pred = model.predict(X_test_scaled)
print(f"Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```